

ME537 Group 1

A1: Product Idea Pitch

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Initial 10

Dimming Candle Holder

Customizable Lamp
Shade

Modular Desk Organizer

Compact Wallet

BookWorm (Light and
bookmark)

Sanitary Portable Handle

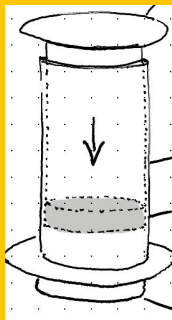
Car Phone Mount

Coffee Press

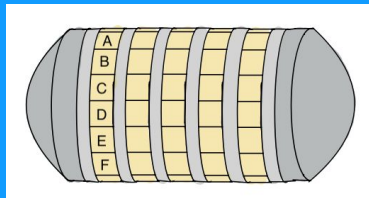
Cryptex Puzzle

High Strength GriGri

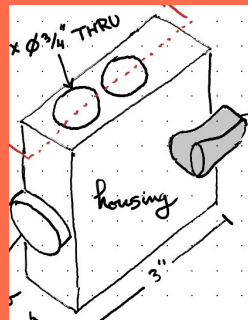
Top 3



The PressMate



Cryptex Puzzle



Boat Gri Gri

The PressMate



3+ parts, small, dimensional fit

- Parts: Plunger, Vessel, Filter, and O-ring
- Tight fit ensuring watertight seal under pressure



2+ mfg processes and material selection

- Aluminum and thermoset rubber
- Sand casting, CNC Lathe and CNC Mill



1+ tool or fixture

- Sand Cast Mold, fixturing inside Mill



Secondary process

- Polishing aluminum interior
- Anodizing body



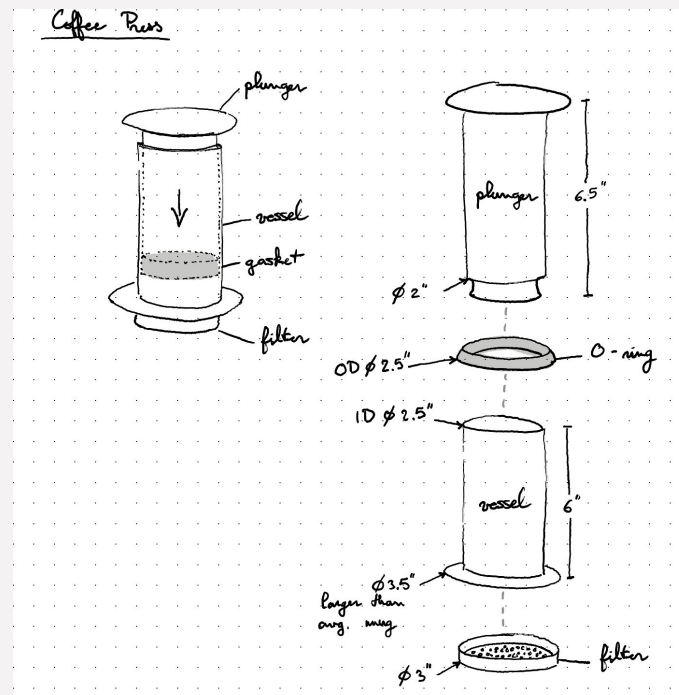
Only ONE purchased part

- Rubber O-Ring



Manufacturing Filter, Watertight Seal

Description: Compact coffee-making device that can create packaged coffee with ease. The built in metal filter removes the ground from the brewed coffee, resulting in a compact and easy-to-use device



Cryptex Puzzle



3+ parts, small, dimensional fit

- Parts: Outer/inner cores, dials, spacers, locking ring, set screw
- Tight fit required for locking function



2+ mfg processes and material selection

- Aluminum and Nylon
- Sand casting, Injection Molding, CNC Lathe/Mill



1+ tool or fixture

- Sand Cast mold, Injection die, fixturing inside mill



Secondary process

- Polishing metal components



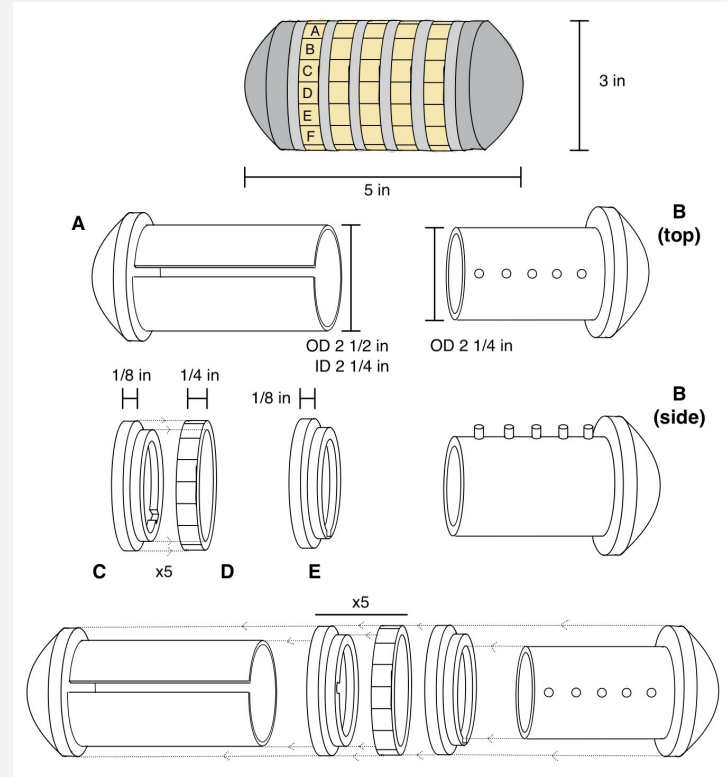
Only ONE purchased part

- Set Screw



Locking mechanism functionality

Description: A portable lockbox popularized by Dan Brown's novel *The Da Vinci Code*. This small, portable puzzle only opens when the dials are turned to their designated positions.



Boat Gri Gri



3+ parts, small, dimensional fit

- Parts: Body, rope lock and release, spring, and set screw.
- Fit ensuring rope security under high forces



2+ mfg processes and material selection

- Steel elements made using CNC Mill and CNC Lathe



1+ tool or fixture

- Fixturing pieces inside the mill



Secondary process

- Plating on components



Only ONE purchased part

- Premade Spring



Corrosive environment, Spring Strength

Description: Assists boat operators tie their boats to pilings and cleats without the need for vast knot knowledge. Allows rope to be permanently secured via a set screw and released with ease.

